

[CH6] The Maturing of the Mainframe: The Rise of IBM

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- In 1957 the film Desk Set came out. The film was a great feat of product placement by IBM and proved to be a great ad for both its products and the company. It portrayed computers as a means to reduce drudgery, not humans.

The IBM 1401

- It was not so much a computer as a computer system
- IBM realised that customers were more interested in solutions to business problems than in the "technical merits of computer design"
- The 1401 came from a need to create a transistor version of the tube-based Model 650 MDC
- Although the 650 had sold more than all the other mainframes put together, most companies were sticking with the Punched card machines for a few reasons:
 - Cost: For the price of an MDC, one could rent a whole array of punched card machines that would do the job just as well
 - Tubes: they were far less reliable than traditional electromechanical machines
 - Programming: Few companies had the funds to hire professional programming staff and IBM offered little software support at the time
 - Peripherals: most 650 peripherals were lackluster products derived from existing punched card machines
- The 1401 used transistors instead of tubes and core memory instead of the drum, which gave it an order-of-magnitude improvement in speed and reliability
- The peripherals needed to give it a decisive advantage over the 650
- To solve the programming issue, IBM created the RPG (Report Program Generator) programming system, which allowed people familiar with accounting machine plugboards to write their own programs after just a few days of training
- IBM developed entire program suites for the customers who didn't want to be involved in programming
- IBM was so far ahead that there was little to no hope for other companies to challenge it.

Revolution, not Evolution: System/360

- IBM was developing 7 different systems. Each needed its own parts, production lines, sales force, programs, etc. This was rapidly becoming overwhelming.
- It started a massive company-wide 5-billion USD project to replace its entire product line with a lineup of universal, compatible computers, complete with software and peripherals.
- The release of the 6 new computers and 44 new peripherals was an incredible success and the system/360 fueled IBM for the next 30 years

Competition strikes back

- IBM's marketing was astonishing but, from a technical and engineering viewpoint, it was just competent
- RCA produced a range of System/360 compatible computers that had superior performance and announced it Spectra 70 range in December 1964, 8 months after the System/360
- Honeywell created a range of computers that were compatible among themselves but not with the System/360
- Control Data recognized that IBM's lineup had no really big computers and dropped regular mainframes altogether in favour of giant number-crunching computers
- General Electric recognized that IBM had no time sharing and became the world leader in time-

- shared computer systems
- UNIVAC continued to compete in airline reservation systems without any major new machine announcements
- Burroughs and NCR built upon their relationship with bankers and retailers
- During the first computer recession of 1970-1971, RCA and General Electric decided to give up their computer divisions.
- From then on, the industry became IBM and the BUNCH (Burroughs, UNIVAC, NCR, Control data, Honeywell)

Systems Men & Management Information Systems

- Computers were not being used by faceless corporations, but by individual people
- No business or government manager buys new technology simply because it is cheaper, faster or better. It takes an analysis of risk, benefit, functionality, cost and reliability, as well as personal, professional and political considerations
- The first application of electronic computing was routing administration (payroll, billing, report generation). It had all already been partially mechanized through electromechanical machines
- This work was delegated to data-processing departments. The machines IBM introduced in the late 1950s were designed to appeal to these departments
- Systems Men were people tasked with developing systems of management that eliminated inefficiency and waste at all levels. For them, the computer was the ideal tool. They introduced the MIS (management information system)
- The idea behind MIS is that computer and communications technologies could create a complete image of the total information environment of a complex and distributed organization.
- Few companies had the resources to implement even part of this extremely expensive goal, but it was appealing nonetheless (except to the middle managers who would be replaced by computers)

The Long March of Computerization

- The rapid spread of electronic computing in the 60s was enabled not just by innovations but also by the standardization that occurred around IBM. This resulted in less competition but made it easier for firms to plan for the future
- Computer manufacturers had a certainty that by aligning themselves with IBM they would have a large market
- Businesses could focus less on whom to buy the computer from and more on what to do with it
- With this wide spread came people who had the expertise and desire to computerize. Management consulting firms were aggressively promoting computerization.
- Firms emerged to develop applications software